CUONG PHAM (Phạm Anh Cường)

◇ Address: Abu Dhabi, UAE ◇ Email: phamanhcuongvna2511@gmail.com ◇ Phone: (+971) 58 520 6140
◇ Webpage: pacman-ctm.github.io ◇ LinkedIn: linkedin.com/in/anhcuong-pham ◇ GitHub: github.com/pacman-ctm

Machine Learning researcher focused on **efficiency** and **optimization** for deep learning.
Research interests: continual learning, parameter-efficient fine-tuning of LLMs, collaborative/federated learning, optimization

EDUCATION

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), UAE — 2024 - 2026
Master of Science in Machine Learning — GPA: 3.72/4.0 ($\sim$ 94%) [Transcript]

- Selected courses: Mathematics for AI, Advanced Machine Learning, Probabilistic & Statistical Inference, Optimization.
- *Thesis*: "Learning in the Null Space: Small Singular Values for Continual Learning".
- Supervisors: Prof. Samuel Horváth and Prof. Praneeth Vepakomma (informally).

VNU University of Engineering and Technology (VNU-UET), Vietnam — 2019 - 2023
Bachelor in Information Technology (Honors Program) — GPA: 3.71/4.0 (8.78/10) [Transcript]

- Graduated with High Distinction, top 5% of CS Department.
- *Thesis*: "Building a Vietnamese image captioning dataset and developing a visual-text transformation model for generating image captions" (9.5/10).
- Supervisors: Prof. Quang-Thuy Ha, Dr. Van-Quang Nguyen, MS. Thi-Hong Vuong.

PUBLICATIONS

(P = preprint, C = conference, O = ongoing paper)

[O.6] **Cuong Anh Pham**, Praneeth Vepakomma, Samuel Horváth. (Ongoing project on efficient continual fine-tuning)

[C.5] **Cuong Anh Pham**, Praneeth Vepakomma, Samuel Horváth. "Learning in the Null Space: Small Singular Values for Continual Learning"
(CPAL '26 - The Third Conference on Parsimony and Learning, **Oral Presentation**).

[P.4] Farshed Abdukhakimov, **Cuong Anh Pham**, Samuel Horváth, Martin Takáč, Slavomir Hanzely. "Polyak Stepsize: Estimating Optimal Functional Values Without Parameters or Prior Knowledge"
(preprint 2508.17288, under-review draft).

[P.3] Quang P.M. Pham, Khoi T. N. Nguyen, Nhi H. Doan, **Cuong A. Pham**, Qinbo Sun, Weimin Qi, Kentaro Inui, Dezhen Song. "SmallPlan: Leverage Small Language Models for Sequential Path Planning with Simulation-Powered, LLM-Guided Distillation"
(preprint 2505.00831).

[P.2] **Anh-Cuong Pham**, Van-Quang Nguyen, Thi-Hong Vuong, Quang-Thuy Ha. "KTVIC: A Vietnamese Image Captioning Dataset on the Life Domain"
(preprint 2401.08100, short version of Bachelor's thesis).

[C.1] Quynh-Trang Pham Thi, **Anh-Cuong Pham,** Ngoc-Huyen Ngo and Duc-Trong Le. "Memory-Based Method using Prototype Augmentation for Continual Relation Extraction"
(IEEE RIVF '22).

EXPERIENCE (Research)

MBZUAI, Abu Dhabi, UAE
Research Engineer (incoming) — (starting 06/2026)

- **Project:** Collaborative learning using limited private samples per client device
  Supervisors: Prof. Praneeth Vepakomma
  Research focuses on efficient collaborative learning and privacy-aware training under data-constrained settings.

Graduate Research Assistant — (10/2024 - 05/2026)

- **(Thesis) Project:** Efficient continual training/fine-tuning with orthogonal-based methods using low-rank updates
  Supervisors: Prof. Samuel Horváth and Prof. Praneeth Vepakomma
  Developing orthogonality-based continual learning methods for preserving prior-task performance with LoRA-style weight updates. Current work focuses on efficient continual fine-tuning (coauthor paper [C.5], ongoing work [O.6]).
- **Project:** Parameter-free SGD Optimization using Polyak stepsize with twin iterates
  Supervisors: Prof. Samuel Horváth and Prof. Martin Takáč
  Contributed to the development of parameter-free Polyak-style adaptive stepsize methods for SGD optimization using twin iterates, aiming to reduce hyperparameter sensitivity during training (coauthor paper [P.4]).

- **(Summer Intern) Topic:** Federated fine-tuning optimization with efficient LoRA updates
  Supervisors: Prof. Praneeth Vepakomma and Prof. Samuel Horváth
  Studied federated optimization and efficient adaptation methods for collaborative learning systems, including reproducing and analyzing techniques such as Fed-SB and LoRA-SB.
- **Project:** Applications of small language models for edge-device robot learning (MBZUAI course project extension)
  Worked on efficient small language model (SLM) approaches for edge-device robotic path planning through simulation-guided distillation from LLMs (coauthor paper [P.3]).

DSKT Laboratory at VNU-UET, Hanoi, Vietnam — 09/2021 - 09/2023
Student Research Intern

- Developed a Vietnamese image-captioning dataset and lightweight Transformer-based captioning models for low-resource vision-language settings (coauthor paper [P.2]).
- Contributed to prototype augmentation methods for continual relation extraction to mitigate catastrophic forgetting with reduced memory storage requirements (coauthor paper [C.1]).

EXPERIENCE (Industry)

Viettel Networks, Hanoi, Vietnam — 10/2023 - 07/2024
Data Scientist

- Researched and applied **Computer Vision** (image processing, segmentation, classification) for tasks with telecommunication images, automating processes for network maintenance.
- Solved **Big Data** problems with large-scale telecommunication data (100M to 10B records), optimizing and automating processes and improving network users' experience by allocating network bandwidth using predicted users' usage.

GHTK Joint Stock Company, Hanoi, Vietnam
R&D Engineer — 08/2022 - 08/2023

- Researched and developed (team of 2 persons) a mail spam filtering, achieving the f1-score of 90% after training with 50k emails, it was then deployed to the company's internal email system.

R&D Engineer Intern — 05/2022 - 08/2022

- MasterDev Season 4 Internship Program in Research and Development domain. Final Project about building a text classification for Vietnamese documents using doc2vec.

INVITED TALKS

"**Learning in the Null Space: Small Singular Values for Continual Learning**"

- CPAL 2026, hosted by ELLIS Tübingen & MPI-IS & Tübingen AI Center (Tübingen, Germany) — 2026
- RelationalML Group at CISPA, hosted by Prof. Rebekka Burkholz (online) — 2026

HONORS & AWARDS

Oral Presentation at The Third Conference on Parsimony and Learning, Tübingen, Germany	2026
Full scholarship for 2 years of the Master of Science program at MBZUAI	2024
(3×) Third Prize (ranked 5th, 6th, 5th) in Linear Algebra, VNU-UET Mathematical Olympiad	2022, 2023, 2024
Yamada Scholarship (6 students/year at VNU-UET)	2022
Excellent Scholarship for top highest GPA from VNU-UET (rank 5th)	2021
Attended Vietnam Mathematical Olympiad for high school students (resulted among top 250/600 students)	2019

CONFERENCES, SCHOOLS & SEMINARS ATTENDED

Efficient Machine Learning Seminar at MBZUAI (Abu Dhabi, UAE)	2025-present
(CPAL '26) The Third Conference on Parsimony and Learning (Tübingen, Germany)	2026
(MLWS '26) MBZUAI Machine Learning Winter School (Abu Dhabi, UAE)	2026
(ICOMP '25) The International Conference on Computational Optimization (Abu Dhabi, UAE)	2025
(ASCOMP '25) The AI Autumn School of Computational Optimization (Abu Dhabi, UAE)	2025

SKILLS

Programming: Python, Java, C/C++
Tools: Microsoft Office, LaTeX, Git, Linux.
Frameworks: PyTorch, Scikit-learn.
Language: English (IELTS 7.0), Vietnamese (native).

REFERENCES

Dr. Samuel Horvath, Assistant Professor at MBZUAI (UAE): (samuel.horvath@mbzuai.ac.ae)
Dr. Praneeth Vepakomma, Assistant Professor at MBZUAI & Visiting Professor at MIT IDSS: (vepakom@mit.edu)